Jiayu Zhou

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Education

Columbia University — New York, NY
M.S. in Computer Engineering; research in robot learning advised by Prof. Yunzhu Li — *May 2027 (Expected)*

University of Illinois Urbana-Champaign — Champaign, IL
B.S. in Computer Engineering, The Grainger College of Engineering — *May 2025*

Zhejiang University — Hangzhou, China
B.E. in Electrical and Computer Engineering, ZJU-UIUC Institute (dual-degree program) — *Jun 2025*

Skills

- **Languages & Tools**: Python, C/C++, CUDA, SQL, Bash, Git, Docker, Slurm, Linux, AWS, Unity
- **Frameworks**: PyTorch, Hugging Face, LLaMA-Factory, Verifiers, vLLM, LangChain, MCP, MuJoCo, Isaac Sim
- **ML / Robotics**: LLM Post-Training (SFT, RL), Agentic RL, RAG, World Models, Sim2Real, Imitation Learning, VLA

Publications

- CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification. *COLM 2026*.
- ToolTurn: Graph-Guided Data Synthesis for Multi-Turn Tool Learning with Verifiable Rewards. *IEEE BigData 2026*.
- Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey. *TMLR 2026*, under review.
- User-Aware Prefix-Tuning is a Good Learner for Personalized Image Captioning. *PRCV 2023*.

Professional Experience

OpenPie (πDataCS: PieCloudDB, PieCloudVector, PieAIStudio) — Shanghai, China
Software Engineer Intern, LLM Applications — cloud-native data computing startup — *Jul 2024 – Aug 2024*

- Built a **government-policy question-answering RAG assistant from 0 to 1** (Hangzhou startup-subsidy policies): Tesseract OCR ingestion, **semantic double-pass chunking** with enumeration-aware merging, **bge-m3** embeddings in an **HNSW** index on **PieCloudVector**, and top-k grounding of **Qwen2-7B-Instruct** with clause-level citations
- Created a **120-question multiple-choice QA (MQA) benchmark** with document-grounded distractors; lifted accuracy from **20.0%** (fixed-width chunking, Llama-2-7B) to **83.3%** via semantic chunking (+30pp), bge-m3 + reranking (+18pp), and model swap (+15pp)
- Deployed as a Dockerized cloud service; post-launch cut **p95 latency 6.1 s to 1.8 s** (cached embeddings, batched vLLM) and **hallucinated answers 23% to 4%** on a 200-query audit via out-of-scope refusal with citations
- Mentored 2 students through the full RAG lifecycle in OpenPie's AI4AI pair-programming camp

Project Highlights

CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification — Remote
Research Assistant with Prof. Philip S. Yu, University of Illinois Chicago; COLM 2026 — *Jan 2026 – Apr 2026*

- Designed a **co-evolutionary framework** where a Skill Generator and an isolated **Surrogate Verifier** iterate generate–verify–refine cycles to build multi-file agent skills without ground-truth tests: **71.1% pass rate on SkillsBench**, **+40.5pp** over no-skill and **+17.6pp** over human-curated skills, beating 5 baselines
- Built the containerized evolution/evaluation harness (Claude Code, Codex, Terminus-2; parallel workers) and ran the **cross-model transfer study**: skills evolved by Claude Opus 4.6 lifted **6 LLMs from 5 providers** (GPT-5.2, Sonnet/Haiku 4.5, Qwen3-Coder, DeepSeek V3, Mistral Large 3) by **+35–44pp**
- Implemented 3 baselines (autonomous skill-creator, one-pass and CoT-guided self-generation) and ablations: the verifier alone adds **+30.0pp**; **4.1 verification cycles vs. 2.4 oracle rounds** per task show the surrogate absorbs ∼40% of iterations; self-evolved skills beat human-curated ones in **9 of 11 domains**

ToolTurn: Multi-Turn Tool-Use Agent RL with Graph-Guided Data Synthesis Remote
Research Assistant with Prof. Philip S. Yu, University of Illinois Chicago; IEEE BigData 2026 *Aug 2025 – Present*

- Designed a **Tool-Dependency Graph** backend that links tools across domains and auto-synthesizes and filters **multi-turn tool-call trajectories on 14 benchmark tasks**, coupling data generation with reward design
- Reused the backend as a **sandbox** that emits **per-turn verifiable rewards** from state matching, replacing the sparse end-of-episode reward and improving credit assignment across turns
- Trained **Qwen2-7B** and **Qwen3-8B** with SFT (LLaMA-Factory) then **GRPO/DAPO + curriculum learning** via Verifiers on 8×A100; $\sim$**10%** gain over the SFT baseline on **BFCL** and **API-Bank** multi-turn, beating all baselines

DARPA TIAMAT Challenge: Sim-to-Sim Transfer for Boston Dynamics Spot New York, NY
Perception lead (1 of 4 Columbia members), advised by Prof. Yunzhu Li *Dec 2025 – Feb 2026*

- Owned perception for TIAMAT: train a **Boston Dynamics Spot** in low-fidelity **Habitat-Sim**, then transfer within a fixed budget to an unseen high-fidelity **Isaac Sim**; Docker-based code passing **smoke and stress tests**
- Designed a real-time **open-set detection dual system**: YOLO-World open-vocabulary proposals verified by **DINOv2** closed-set matching, **0.97 precision on unseen objects**; benchmarked SAM3 and DINOtxt alternatives
- Split DINOv2 into precompute and verify-and-correct stages, **overlapping precompute with YOLO inference** to cut round-trip latency **780 ms to 620 ms** (batch 8, 1×L40S); served to navigation (VLFM) and locomotion

Augmenting Simulation with World Models for Sim2Real Style Transfer New York, NY
Lead author (in progress), advised by Prof. Yunzhu Li, Columbia University *Jun 2026 – Present*

- Thesis: a simulator augmented by a **style-transfer world model** is both a trustworthy evaluator and a scalable training-data engine; testbed: **2.5 mm-tolerance gear assembly** in **MuJoCo (SDF collision)** on a **Franka FR3**
- Collected **40+ real demos (20 min)**; used Canny-edge, depth, and GroundingDINO + SAM2 + CLIP masks as controls to post-train and benchmark **Cosmos 3 Nano**, **Cosmos Transfer 2.5**, **SDXL**, and **Img2Img-Turbo**; policy success in augmented sim correlates with real-robot success at **Pearson 0.93** with matching failure modes
- Two sim-only training pipelines: (a) **512 offline sim demos + BC + DAgger / residual RL**; (b) **curriculum PPO teacher** on privileged state + **augmented on-policy distillation (OPD)** into a vision student, boosting success to $\sim$**100%** vs. $\sim$80% for an ACT policy trained on real demos
- Accelerated style-transfer inference **8.4×** (Cosmos 3 Nano on GH200, **2.8 s to 0.33 s/frame**) via **DMD2 distillation** + grafted control branch, **context parallelism**, a vllm-omni warm server, and **hierarchical sparse attention**

Interactive World Model for VLA Imitation Learning and RL Post-Training New York, NY
Research Assistant, advised by Prof. Yunzhu Li, Columbia University *Feb 2026 – May 2026*

- Trained a task-specific **interactive world model** on $\sim$**4 h of real-robot data** that rolls out physically consistent, **long-horizon (10+ min)** trajectories in **real time (15 FPS on one RTX 4090)**
- Designed a **closed-loop data engine** around the model: sample expert trajectories for imitation, warm-start **MPPI** optimization, fold optimized trajectories back into training, and replay RL failures on the real robot
- Post-trained **Diffusion Policy** with **SAC** using **VLM-logit rewards** on PushT and mug grasping inside the world model; evaluated policies in both the real world and the model to quantify its fidelity as an evaluator

FuncGrasp: Semantically-Guided Functional Dexterous Grasping New York, NY
Course project, advised by Prof. Yunzhu Li, Columbia University *Aug 2025 – Dec 2025*

- Built a **VLM + diffusion** grasping stack for the **Shadow Hand**: LoRA-tuned **Qwen2-VL-2B** maps language, RGB-D, **point clouds (PointTransformer)**, and state to per-subtask contact maps; a **DDPM** conditioned on keyframe contact maps predicts 34-D action chunks
- Extended **DexGraspNet 1.0** with affordance maps, contact points, and multi-view images in MuJoCo; added **collision losses (ERF, SPF, SRF)**, improving success and semantic following by $\sim$**31%** on DexGraspNet and RealDex

EmbRAG: LLM Reasoning over Incomplete Temporal Knowledge Graphs Champaign, IL
Research Assistant, advised by Prof. Hanghang Tong, University of Illinois Urbana-Champaign *Sep 2023 – Jan 2025*

- Tackled graph incompleteness and rule ambiguity: sampled rules by **random walk**, scored confidence by hit frequency and temporal association, and trained a decoder to generate new rules from high-confidence prompts
- Filled missing adjacency entries with **fuzzy logic + rule-embedding similarity**, enabling **end-to-end LoRA fine-tuning** of the LLM; reached **SOTA on multiple tKG benchmarks**

Senior Design: VR Teleoperation of a UR3e Arm for Object Grasping Hangzhou, China
Advised by Prof. Liangjing Yang, Zhejiang University; open-sourced *Jan 2025 – Jun 2025*

- **Real-time remote control (latency $<$ 33 ms)** of a **UR3e** via a Unity digital twin and Meta Quest 3S VR interface; 3D-printed gripper with **closed-loop force control (STM32 + CAN)** on a custom PCB